

Data Governance as the Foundation of AI Systems

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Lecture 8

The Real Gap

The primary limitation in AI adoption is governance, not capability.

- ▶ AI capabilities are rapidly advancing.
- ▶ Tools for generation are widely available.
- ▶ Startups prioritize capability over integration.
- ▶ Enterprise adoption remains slow.
- ▶ Barriers are organizational.
- ▶ Governance is the missing layer.

AI as Data Processing

AI systems should be understood as data processing utilities.

- ▶ Inputs are data.
- ▶ Outputs are transformed artifacts.
- ▶ Models operate on structured and unstructured data.
- ▶ Value depends on data quality.
- ▶ Integration determines usability.
- ▶ Processing implies governance obligations.

Core Principle

Systems that process data must comply with governance.

- ▶ AI systems ingest organizational data.
- ▶ They generate derived data.
- ▶ They redistribute information.
- ▶ They interact with sensitive datasets.
- ▶ They influence decisions.
- ▶ They fall under governance frameworks.

What Is Data Governance

Data governance defines how data is managed.

- ▶ Establishes policies.
- ▶ Defines accountability.
- ▶ Assigns decision rights.
- ▶ Ensures compliance.
- ▶ Maintains quality.
- ▶ Aligns with strategy.

What Is Data Management

Data management implements governance.

- ▶ Handles data collection.
- ▶ Manages storage systems.
- ▶ Maintains integrity.
- ▶ Supports access workflows.
- ▶ Enforces security controls.
- ▶ Executes lifecycle operations.

Governance vs Management

Governance defines rules while management executes them.

- ▶ Governance sets expectations.
- ▶ Management applies rules.
- ▶ Governance is strategic.
- ▶ Management is operational.
- ▶ Alignment is required.
- ▶ Separation clarifies roles.

Enterprise Constraints

Organizations prioritize stability over innovation.

- ▶ Data security is critical.
- ▶ Legal compliance is required.
- ▶ Integration is necessary.
- ▶ Auditability is expected.
- ▶ Reliability is essential.
- ▶ Control is mandatory.

Enterprise Pain Points

Adoption is limited by organizational complexity.

- ▶ Risk of data exposure exists.
- ▶ Departments may conflict.
- ▶ Legacy systems complicate integration.
- ▶ Ownership is unclear.
- ▶ Policy enforcement is difficult.
- ▶ Resistance is common.

Key Insight

Governance determines adoption rather than technical quality.

- ▶ Better models do not ensure use.
- ▶ Security overrides innovation.
- ▶ Compliance blocks deployment.
- ▶ Integration determines value.
- ▶ Trust drives acceptance.
- ▶ Governance determines viability.

Why Agentic AI Fails

Agentic systems often ignore governance requirements.

- ▶ They assume open data access.
- ▶ They lack control mechanisms.
- ▶ They omit audit trails.
- ▶ They cannot ensure compliance.
- ▶ They operate opaquely.
- ▶ They ignore constraints.

Common Failures

Typical issues reflect governance gaps.

- ▶ Sensitive data enters prompts.
- ▶ Outputs lack traceability.
- ▶ Decisions lack accountability.
- ▶ Logs are incomplete.
- ▶ Results are not reproducible.
- ▶ Oversight is missing.

Opaque Decisions

Lack of transparency undermines trust.

- ▶ Reasoning is hidden.
- ▶ Outputs are unexplained.
- ▶ Steps are not recorded.
- ▶ Auditing is difficult.
- ▶ Errors are untraceable.
- ▶ Trust declines.

Data Lifecycle Overview

Governance spans the entire data lifecycle.

- ▶ Plan data usage.
- ▶ Create or obtain data.
- ▶ Store and maintain data.
- ▶ Use data operationally.
- ▶ Enhance through analysis.
- ▶ Dispose when unnecessary.

AI and Lifecycle

AI participates in every stage of the lifecycle.

- ▶ Prompts are inputs.
- ▶ Artifacts store transformations.
- ▶ Outputs become assets.
- ▶ Logs capture activity.
- ▶ Training data influences behavior.
- ▶ Governance applies everywhere.

Plan Stage

Planning defines system boundaries.

- ▶ Identify data sources.
- ▶ Define use cases.
- ▶ Assign ownership.
- ▶ Establish compliance.
- ▶ Set boundaries.
- ▶ Align with goals.

Create or Obtain

Data enters through multiple channels.

- ▶ User prompts provide data.
- ▶ APIs add inputs.
- ▶ Artifacts accumulate.
- ▶ Validation ensures consistency.
- ▶ Metadata is captured.
- ▶ Provenance is tracked.

Store and Maintain

Data must be securely managed.

- ▶ Outputs require storage.
- ▶ Access must be controlled.
- ▶ Encryption protects data.
- ▶ Retention policies apply.
- ▶ Integrity is maintained.
- ▶ Documentation is required.

Use Stage

Data is used to generate value.

- ▶ Systems transform inputs.
- ▶ Decisions are made.
- ▶ Access must be restricted.
- ▶ Usage must be authorized.
- ▶ Outputs must be validated.
- ▶ Compliance is enforced.

Enhance Stage

Data is refined and improved.

- ▶ Outputs are corrected.
- ▶ Patterns are identified.
- ▶ Errors are addressed.
- ▶ Feedback improves systems.
- ▶ Quality is monitored.
- ▶ Governance ensures alignment.

Dispose Stage

Data must be removed appropriately.

- ▶ Retention schedules apply.
- ▶ Legal rules may override deletion.
- ▶ User requests trigger removal.
- ▶ Secure deletion is required.
- ▶ Disposal is tracked.
- ▶ Compliance is ensured.

Governance Requirements

AI systems must meet governance constraints.

- ▶ Enforce access control.
- ▶ Maintain logs.
- ▶ Enable traceability.
- ▶ Validate behavior.
- ▶ Apply retention rules.
- ▶ Support audits.

Security Connection

Governance directly supports security.

- ▶ Prevents data leakage.
- ▶ Limits exposure.
- ▶ Reduces attack risk.
- ▶ Ensures accountability.
- ▶ Supports incident response.
- ▶ Aligns with regulation.

Strategic Opportunity

Governance creates competitive advantage.

- ▶ Many ignore governance.
- ▶ Solutions fail enterprise needs.
- ▶ Safe AI is required.
- ▶ Compliance-ready systems are rare.
- ▶ Integration-focused tools are valuable.
- ▶ Governance differentiates.

Market Positioning

Successful systems align with enterprise requirements.

- ▶ Emphasize compliance.
- ▶ Integrate with infrastructure.
- ▶ Provide transparency.
- ▶ Control data flows.
- ▶ Align with governance.
- ▶ Deliver trusted outputs.

Organizational Alignment

AI requires cross-functional coordination.

- ▶ Security ensures protection.
- ▶ Legal ensures compliance.
- ▶ Governance defines policy.
- ▶ Product defines use cases.
- ▶ Engineering builds systems.
- ▶ Coordination is required.

Organizational Failure

Late governance involvement causes failure.

- ▶ Security is consulted too late.
- ▶ Compliance issues emerge late.
- ▶ Systems are rejected.
- ▶ Rework is required.
- ▶ Projects stall.
- ▶ Early alignment is critical.

Core Synthesis

AI systems must be governed from the start.

- ▶ Governance is not optional.
- ▶ It must be architectural.
- ▶ Design reflects constraints.
- ▶ Data flows are controlled.
- ▶ Outputs are accountable.
- ▶ Systems are auditable.

Series Integration

Governance integrates all prior phases.

- ▶ Model behavior is understood.
- ▶ Systems are structured.
- ▶ Failures are identified.
- ▶ Risks are analyzed.
- ▶ Governance unifies phases.
- ▶ Systems thinking dominates.

Final Insight

AI success depends on systems, not models.

- ▶ Models are one component.
- ▶ Systems define behavior.
- ▶ Data defines value.
- ▶ Governance defines viability.
- ▶ Organizations drive adoption.
- ▶ Integration ensures success.

Closing Statement

AI is fundamentally a governance and systems problem.

- ▶ Technology alone is insufficient.
- ▶ Structure determines reliability.
- ▶ Governance ensures safety.
- ▶ Alignment enables adoption.
- ▶ Strategy drives impact.
- ▶ Governed systems define the future.

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